

Rakshit Ranka

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Department of Computer Science and Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute/School	CPI/%
2024 - Present	B.Tech	Indian Institute of Technology, Kharagpur	9.16/10
2024	CBSE (XII)	RMG Maheshwari English School, Surat	95%
2022	CBSE (X)	RMG Maheshwari English School, Surat	97%

Awards and Achievements

- Secured **All India Rank 258** in the **Joint Entrance Exam (Advanced)** 2024 among 1,80,000+ candidates,conducted by IIT Madras
- Secured **All India Rank 735** in the **Joint Entrance Exam (Mains)** 2024 among 14,00,000+ candidates,conducted by the NTA
- 1 of only 4 students from IIT Kharagpur to be shortlisted for the **Aditya Birla Scholarship 2024** for academic and overall excellence
- Qualified **RMO 2023** and was among 41 students from Gujarat invited for the Indian National Mathematical Olympiad (**INMO**) 2024
- Awarded a **Certificate of Merit** after successfully clearing the cutoff in the Indian Olympiad Qualifier in Mathematics (**IOQM**) 2021
- Placed in **state top 1%** in NSEP 2022 and **national top 1%** in 2023; qualified for the Indian National Physics Olympiad (**INPhO**) 2024
- Secured a position in the **national top 1%** in NSEA 2023, thus qualifying for the Indian National Astronomy Olympiad (**INAO**) 2024
- Ranked among the **national top 1%** in NSEC 2023 to successfully qualify for the Indian National Chemistry Olympiad (**INChO**) 2024
- Cleared the Merit Index and was in the **national top 1%** in NSEJS 2020; qualified for the Indian National Junior Science Olympiad 2021

Competition/Conference

- Attained the title of **Expert** with a peak rating of **1706** on Codeforces, a competitive programming platform (Handle : rakshit_ranka)
- Obtained Global Rank **510** in Educational Codeforces Round 190 (Div. 2) and Global Rank **812** in Codeforces Round 1091 (Div. 2)
- Achieved University **Rank 1** and Global Rank 567 in Stage 1 of the International Quant Championship 2025 by organised by WorldQuant
- Placed **564th** overall in Round 2 of the **Meta Hacker Cup 2025** among a pool of more than 5,000 participants who cleared Round 1
- Secured rank **152** in the Prior round of **AMS Derive 2026** and qualified for the Posterior round among more than 2,500 registrants

Key Projects

- OORT - An Astrophysical Simulation DSL** 🌀 | Course Project | OCaml, Python, Pygame Feb'26 – Apr'26
Software Engineering Lab Project under Prof. Jibesh Patra and Prof. Soumyajit Dey
 - Developed an astrophysics domain-specific language,using **OCaml** for the **tokenizer**, **parser**, and **semantic analyzer** for type checking
 - Designed the **Python** code generator, translating validated ASTs into interactive **Pygame** simulations via setup/game-loop split
 - Implemented the **n-body physics engine** using **velocity Verlet integration**, with inelastic collisions conserving mass and momentum
 - Automated build and test workflows via **GitHub Actions**, integrating **Dune**, **OUnit2**, & **Merlint** for continuous code quality checks
- CHIP-8 Emulator** 🌀 | Self Project | C++, SDL3 Apr'26 – May'26
 - Built a CHIP-8 emulator in **C++** implementing **34 opcodes**, stack-based subroutines, and hardware timers across 4KB of memory
 - Implemented an efficient instruction dispatch pipeline using nested switch-case statements for **O(1) bitwise opcode decoding**
 - Size-validated ROM loading via binary file streams, and **boundary validation** providing safety against out-of-bounds memory access
 - Designed an **SDL3 rendering** engine with XOR sprite drawing logic, precise collision detection, and **CRT phosphor visual effects**
- Limit Order Book and Matching Engine** 🌀 | Self Project | C++ May'26 – Jun'26
 - Built a **C++** order book and matching engine supporting **Market & Limit** orders across **GTC**, **IOC**, and **FOK** time-in-force policies
 - Maintained **price-time priority** using a **B-Tree based map** for price levels and a **doubly-linked list** for FIFO ordering within levels
 - Achieved **O(1)** order lookup and cancellation via a **flat hash map** and direct linked-list node removal from a **custom memory pool**

Technical Skills

- Programming Languages:** Proficient: C, C++, OCaml, Python, LaTeX | Familiar: SQL, JavaScript, Bash, AWK, Prolog
- Libraries and Tools:** C++ STL, NumPy, Pandas, Matplotlib, SDL3, Git, GitHub, Makefile, GDB, Valgrind, gprof

Coursework Information

Programming & Data Structures* | Switching Circuits & Logic Design* | Algorithms - 1* | Software Engineering* | Systems Programming* | Probability & Statistics | Discrete Structures | Formal Language & Automata Theory | Foundations of AI & ML | Astronomy & Astrophysics | Programming Paradigms for Computation | Signals & Systems | Advanced Calculus | Linear Algebra, Numerical and Complex Analysis

* - Courses accompanied by a lab component

Positions of Responsibility

- Secretary, CodeClub, IIT Kharagpur** Mar'25 – Present
 - Organised CSE Freshers' Nite and farewell for the graduating batch of the CSE department, and selections from the first-year batch
 - Problem setter for Freshers' Codenite 2026, a campus-wide competitive programming contest for the freshers batch of 2025-26
- Associate Member, Grimoire of Code, IIT Kharagpur** Nov'24 – Present
 - Prepared and tested problems, solutions, and an editorial for a contest in the GOC CDC Series 2026, ensuring its smooth conduct
- Strategist, The KGPIan Game Theory Society, IIT Kharagpur** Oct'24 – Apr'26
 - Organised a campus-wide puzzle-based treasure hunt to promote Game Theory and the society's activities among the first-year batch
 - Presented a review of research papers analysing Game Theoretic aspects of the EUR-USD FOREX market and the Ethereum Blockchain

Extra Curriculars

- Member of libbabel.so, a CTF team ranked **16th** in India; specialize in Open Source Intelligence and Digital Forensics challenges
- An avid quizzer; represented IIT Kharagpur at the Inter IIT Cultural Meet 8.0 as part of the **Bronze-winning** Quiz contingent